POLARIZATION AND DISCRIMINATION IN A MULTI-AGENT SYSTEM OF NEURAL NETWORKS

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ABSTRACT

Populations of learning agents are increasingly deployed as societies: agents that read each other's outputs, weigh how far to trust them, and revise their positions accordingly. We ask whether discriminatory group structure can emerge in such a population from the *learning rule alone*, with no biased data and no group-level preference anywhere in the system. We study a society of adaptive agents that each carry a classifier and update, online and optimally, both their opinions and the trust they assign to every other agent, and inject the minimal possible bias: a fraction f_d of agents extend their representation of a message with a feature that carries no information about its content and depends only on the sender's group label, at strength d. The result is a phase diagram in (d, f_d) with four regimes separated by sharp transitions — neutral polarisation unrelated to group membership, two discriminatory phases distinguished by whether hostility still needs disagreement to sustain it, and, for bias favouring the out-group, a frustrated spin-glass regime in which no coherent faction can form. Three findings bear on how such systems should be built and evaluated. Discrimination requires a quorum of biased agents — below $f_d \approx 0.4$ the population is neutral however strong the bias — and past it the transition is abrupt, so a population can be discriminatory while every agent in it passes an individual audit. The effect acts entirely through the sector of the algorithm that estimates how reliable other agents are, so it is a hazard specific to agents that model each other rather than a generic consequence of an irrelevant feature. And the *order* of polarisation is controlled by the complexity of the agenda: below $\alpha = P/K \approx 1.7$ distrust forms first and drags opinions after it, above it opinions separate first. The order parameters we use are defined only through pairwise opinion alignment and pairwise trust, so they are measurable in any population of interacting agents, whatever the agents are made of.

1 INTRODUCTION

Systems built from many interacting learned agents are now routine: agents that debate, review each other's work, negotiate, and simulate populations (Park et al., 2023; Du et al., 2024; Chuang et al., 2024). Each agent reads what others produce, forms some judgement of how much to rely on them, and adjusts its own position accordingly. That is a society, and societies have collective states that no individual chose. The question this paper asks is whether one of the collective states available to such a system is *discrimination*: a population that sorts itself into mutually distrustful groups along a label that carries no information.

The usual explanation for group bias in machine learning locates it in the data: a model trained on a biased corpus reproduces the bias. That account cannot explain a population-level effect, because it says nothing about interaction. We therefore ask a different question. Suppose every agent is individually well-specified — learning by an algorithm that is provably optimal in the setting it was designed for — and suppose the only defect is that some agents attend to one irrelevant feature of a message, namely who sent it. What does the *population* do?

We study this in a model in which the analysis can be carried out: a society of N agents, each carrying a perceptron over a K-dimensional embedding of issues, interacting by stating opinions about issues drawn from a shared agenda of size P. Two features make the model unusually apt for the question. First, agents can agree on some issues and disagree on others, so opinion distance is graded rather than binary. Second, learning is derived rather than posited: the update follows from a Bayesian step on the agent's belief, projected back onto a Gaussian family by maximum entropy (Opper, 1996; 1998; Caticha, 2020), and it carries alongside the weights an explicit, dynamical estimate of how noisy each other agent is as a channel — a quantity we can read as trust (Alves & Caticha, 2016; Caticha & Vicente, 2011). Cognitive-sounding behaviour then appears without being written in: surprise, blame attribution between belief-revision and source-derision, and the ability to learn the *opposite* of what a distrusted source says.

Into this society we inject the smallest bias that can be correlated with a group label. A fraction f_d of agents shift their perceived agreement with a message by $\pm d$ according to whether the sender shares their label. Nothing else distinguishes the groups; the labels are informationally inert; the two groups are treated symmetrically, so any asymmetry that emerges is spontaneous. The resulting collective behaviour, as a function of (d, f_d), is the subject of the paper.

Contributions.

- **A phase diagram of emergent discrimination** (§6): four regimes with sharp boundaries — neutral polarisation unrelated to class, two discriminatory phases distinguished by whether opinion still tracks the group split, and a frustrated spin-glass regime when the bias favours the out-group. Discrimination needs a quorum: below $f_d \approx 0.4$ the society stays neutral however strong the bias, and above it the transition in d sharpens rapidly.
- **A mechanism specific to good learners** (§5): the bias displaces the separatrix of the learning flow, enlarging one dissonance-free basin at the other's expense. It acts only through the algorithm's trust sector, so optimising for the small-group setting is what makes a population vulnerable in the large-group one.
- **Which comes first, distrust or disagreement** (§4): even with no bias at all, the order of polarisation is set by the agenda complexity $\alpha = P/K$, reversing at $\alpha \approx 1.7$. This turns a contested question about polarisation (Fiorina & Abrams, 2008; Iyengar et al., 2012) into a statement about a measurable parameter.
- **Order parameters that do not depend on the substrate** (§4.1). They need only, per pair of agents, a signed opinion alignment and a signed trust — never weights, gradients or the embedding dimension. Two elicited matrices per population therefore suffice to locate any interacting multi-agent system on the diagram, which matters because the per-agent audits that are standard practice cannot see the phase at all.

Our reading is deliberately narrow: these are simple machines, and extending conclusions from them to human societies is not automatic. What the model does support is a claim about learning systems — that an inference rule optimal for a pair of agents, transplanted into a large population, can convert an informationally worthless label into a stable architecture of group distrust. That is a failure mode of the interaction rather than of the training data, and it is invisible to any evaluation that examines agents one at a time.

2 Related work

Societies of language-model agents. Multi-agent language systems have been used to simulate populations and to study what emerges from their interaction: generative agents with memory and social behaviour (Park et al., 2022; 2023), debate procedures that improve factuality (Du et al., 2024), opinion exchange over networks (Chuang et al., 2024), and language models as stand-ins for survey respondents (Argyle et al., 2023; Aher et al., 2023), with systematic biases documented in simulated debate (Taubenfeld et al., 2024). This literature is largely empirical: it reports what a particular population does. What it lacks is a model in which the collective outcome can be *derived* and its phase boundaries located — which is what we supply, along with order parameters cheap enough to measure on any such population.

Bias and fairness in machine learning. The dominant framing locates group bias in data and in single-model decisions, as representational harm (Blodgett et al., 2020; Bender et al., 2021) or as allocative unfairness with formal criteria attached (Dwork et al., 2012; Hardt et al., 2016; Barocas et al., 2023). Our mechanism is orthogonal and complementary: the outcome here is produced by an irrelevant feature interacting with the *learning rule* across a group of agents, and it is undetectable by any audit that inspects agents one at a time, since a single agent's bias is $O(d)$ while the collective state it induces is discontinuous in d. That irrelevant features degrade a classifier is old; that they can restructure a *society* of classifiers is the point here.

Opinion dynamics and social balance. Models of consensus and polarisation in interacting populations run from bounded-confidence and mixing rules (Deffuant et al., 2000; Hegselmann & Krause, 2002) through culture dissemination (Axelrod, 1997) to reinforcement-driven echo chambers (Baumann et al., 2020); see Castellano et al. (2009) for a survey. In most of that work an agent's state is a scalar or a discrete label, so two agents either agree or do not. Representing agents as classifiers over an issue space (Caticha & Vicente, 2011) makes partial agreement possible, which is what lets us ask whether distrust follows disagreement or the reverse. Our second diagnostic, social balance over triples (Heider, 1946; Cartwright & Harary, 1956; Antal et al., 2005; Marvel et al., 2011), is usually applied to sign structures that are themselves the primitive objects; here both sign structures are generated by inference, and their balance is what separates organised bloc formation from disorder — the distinction that identifies the frustrated phase, which pair correlations alone would report as nothing happening.

Optimal on-line learning, and affective polarisation. The learning rule is the optimised on-line algorithm for the student–teacher problem, given a Bayesian reading by Opper (1996) and an entropic-dynamics formulation by Caticha (2020); the identification of the inferred channel-noise level with distrust, and its promotion to a dynamical variable, is due to Alves & Caticha (2016). Appendix F gives the fuller genealogy. Our contribution is not the algorithm but what it does to a population when one class-correlated feature is added. Separately, whether groups come to dislike each other because they disagree or the reverse is contested in political science (Fiorina & Abrams, 2008; Iyengar et al., 2012; Iyengar & Westwood, 2015); we do not adjudicate it, but identify a parameter that selects which occurs.

3 A SOCIETY OF NEURAL NETWORKS THAT LEARN FROM EACH OTHER

3.1 MICROSCOPIC VARIABLES

A society of N agents discusses an agenda of P issues. An issue is a vector $\hat{x} \in \mathbb{R}^K$: the embedding of an assertion in the K dimensions that are relevant to the question at hand. All agents embed issues identically and differ only in what they make of them, so the model separates the representation of an issue from the opinion an agent holds about it.

Each agent carries a single-layer perceptron with weights $w_I \in \mathbb{R}^K$ and states the opinion $\sigma_I = \operatorname{sign}(w_I \cdot \hat{x}) \in \{\pm 1\}$. This is the least structure that lets two agents agree on some issues and disagree on others; a model in which each agent is a single spin cannot do that, and the possibility of partial agreement turns out to drive the collective behaviour we report. Agents also belong to one of two classes, A or B, recorded as $\kappa_I = \pm 1$: a label that is visible to others and carries no information about any issue. Until §5 nothing in the dynamics refers to it.

An agent's state has two sectors. The *opinion* sector is a Gaussian belief over its own weights, with mean $\hat{w}_I$ and covariance C_I. The *trust-attribution* sector is a Gaussian belief, for every other agent J, over how unreliable J is as a source: mean $\mu_{J|I}$ (the *distrust* I assigns to J) and variance $V_{J|I}$. We call it the trust sector for short. Writing the joint belief as a product of Gaussians,

$$Q(w_I, z_I \mid \lambda_I) \;=\; \mathcal{N}(\hat{w}_I, C_I) \prod_{J \neq I} \mathcal{N}(\mu_{J|I}, V_{J|I}), \tag{1}$$

the microscopic variables of the society are $\lambda_I = \{\hat{w}_I, C_I, \mu_{J|I}, V_{J|I}\}$. The means fix what an agent thinks and whom it trusts; the (co)variances fix how firmly.

Certainty and immobility are the same thing. The second moments play a double role worth naming before the update is written down. C_I and $V_{J|I}$ are the agent's uncertainties, but they also multiply every change it makes. They are effective, adaptive learning rates as much as error bars, and in the limit $C_I, V_{J|I} \to 0$ the dynamics freezes: an agent absolutely certain of its opinions and of its trust in everyone cannot be moved by any message, however surprising. That is the Bayesian statement that evidence cannot shift a delta-function prior.

3.2 DYNAMICS

At each step an issue $\hat{x}$, an emitter e and a receiver r are drawn uniformly. The emitter states σ_e; the receiver treats it as a noisy report of an unknown true label, where the noise level is precisely what its distrust of the emitter encodes. A Bayesian step on (1) leaves the exponential family, and projecting the posterior back onto a Gaussian by maximum entropy — relative to the prior, subject to the first two moments — keeps the belief in the form (1) and yields a closed update. This is the entropic-dynamics construction, and it is what makes the algorithm optimal in the small-group (student–teacher) setting rather than merely plausible. Appendix E carries the derivation, including the two modelling choices that keep the algebra Gaussian: taking the probability that the emitter conveys the wrong label to be $\Phi(z)$ with z Gaussian, and letting an inexperienced agent be correspondingly unsure it has parsed the issue as the emitter did.

Two scaled fields carry everything the update depends on:

$$h_w \;=\; \frac{\hat{w}_r \cdot \hat{x}\,\sigma_e}{\gamma_C}, \qquad h_\mu \;=\; \frac{\mu_{e|r}}{\gamma_V}, \qquad \gamma_C = \sqrt{1 + \hat{x}\cdot C_r \hat{x}}, \quad \gamma_V = \sqrt{1 + V_{e|r}}. \tag{2}$$

We call h_w the *opinion field*: it is positive when the receiver already agrees with the message and large when it holds that opinion confidently. We call h_μ the *distrust field*: positive when the receiver distrusts the emitter. In these variables the evidence for the message is

$$Z \;=\; \Phi(h_w) + \Phi(h_\mu) - 2\,\Phi(h_w)\Phi(h_\mu), \tag{3}$$

which is small in two quadrants — agreeing with a distrusted emitter, and disagreeing with a trusted one — and large in the other two. Low evidence is surprise, and surprise is what drives learning. The update is a gradient step in $\log Z$ in both sectors,

$$\hat{w}_r \leftarrow \hat{w}_r + \frac{F_w}{\gamma_C}\,\sigma_e\, C_r \hat{x}, \qquad\qquad C_r \leftarrow C_r + \frac{F_C}{\gamma_C^2}\,(C_r\hat{x})(C_r\hat{x})^\top, \tag{4}$$

$$\mu_{e|r} \leftarrow \mu_{e|r} + \frac{F_\mu}{\gamma_V}\,V_{e|r}, \qquad\qquad V_{e|r} \leftarrow V_{e|r} + \frac{F_V}{\gamma_V^2}\,V_{e|r}^2, \tag{5}$$

with the four modulation functions given by the derivatives of $\log Z$:

$$\begin{aligned} F_w &= \big(1 - 2\Phi(h_\mu)\big)\,\frac{g(h_w)}{Z}, & F_\mu &= \big(1 - 2\Phi(h_w)\big)\,\frac{g(h_\mu)}{Z}, \\ F_C &= -F_w(F_w + h_w), & F_V &= -F_\mu(F_\mu + h_\mu), \end{aligned} \tag{6}$$

where g is the standard normal density. The first update in (4) is a Hebbian step $\sigma_e\hat{x}$ with a tensorial, self-annealing learning rate — the C_r of §3.1 in its learning-rate role.

Three properties of (6) do the work in what follows, and none of them was put in by hand. **Trust gates learning and can invert it**: the prefactor $1 - 2\Phi(h_\mu)$ in F_w is negative when the emitter is distrusted, so a distrusted emitter is not ignored but *anti-learned* from, generating effectively antiferromagnetic couplings. (A comparable reversal is documented in humans, where a norm advertised by a distrusted source can push behaviour the other way, Keizer et al., 2011.) **Agreement builds trust**: symmetrically, $1 - 2\Phi(h_w)$ in F_μ is negative when the receiver agrees, so whether μ rises or falls is decided by the *sign of the perceived agreement* — the hinge that §5 turns. **Only one sector yields at a time**: the sectors are mirror images, $F_w(x, y) = F_\mu(y, x)$ and $F_C(x, y) = F_V(y, x)$, and the surprise is absorbed by whichever field is smaller in magnitude, crossing over at $h_\mu = h_w$ (Figure 1). Meeting a distrusted agent who agrees with you costs you either your distrust or your opinion, whichever you hold less firmly — blame attribution emerging from the inference problem rather than assumed.

4 EMERGENT POLARISATION

Nothing in §3 refers to an agent's class, so before adding any bias we ask what the society of §3.2 does unaided. It polarises. This section says how that is measured, shows it, and then shows that the *order* in which the two sectors polarise is set by a single ratio.

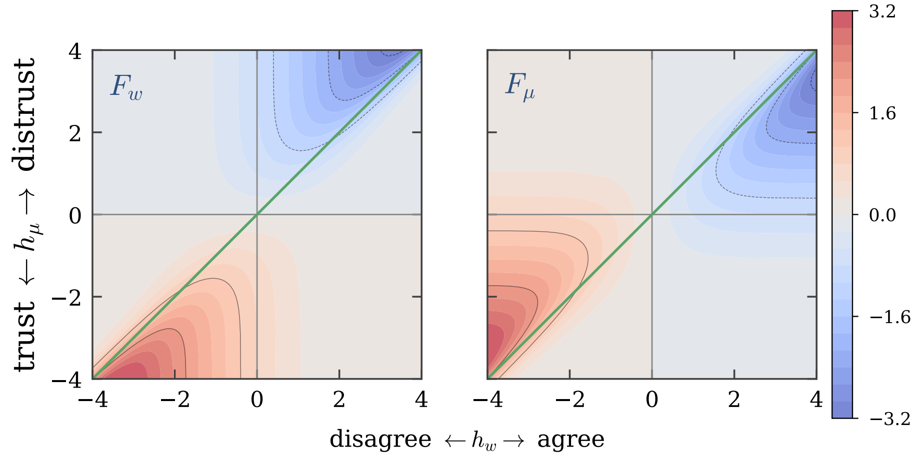

Figure 1: **Learning is driven by dissonance.** The two modulation functions that move the means, over the plane of the opinion field h_w and the distrust field h_μ: F_w, which revises the receiver's opinion, and F_μ, which revises its trust in the emitter. Nothing happens where the message is unsurprising; everything happens in the two dissonant quadrants, agreeing with a distrusted emitter (upper right) and disagreeing with a trusted one (lower left). The two panels are reflections of each other across the green line $h_\mu = h_w$, which is the symmetry $F_w(x, y) = F_\mu(y, x)$ between the sectors, and that line is also where blame for a surprise passes from one sector to the other (Appendix C). The two functions that anneal the uncertainties, F_C and F_V, obey the same symmetry and are shown in Appendix B. Both functions diverge where the evidence goes to zero in the far corners, so the colour scale is clipped at ± 3.2.

4.1 Order parameters

Two quantities are defined per pair of agents rather than for the society as a whole. The *opinion overlap* is the cosine of two weight vectors, $\rho_{IJ} = \hat{w}_I \cdot \hat{w}_J / (\|\hat{w}_I\| \|\hat{w}_J\|)$, equal to $+1$ when two agents agree on every issue and -1 when they disagree on every one. The *trust* a receiver I places in an emitter J is $\eta_{J|I} = 1 - 2\Phi(h_\mu)$, the distrust field mapped onto a bounded, sign-carrying scale, equal to $+1$ for a fully trusted emitter and -1 for a fully distrusted one.

Neither, on its own, says anything about the society: a pair can disagree in a world where everyone disagrees with everyone. What distinguishes organised disagreement from disorder is whether the disagreements are mutually consistent, which is a statement about triples rather than pairs (Heider, 1946; Cartwright & Harary, 1956). A triple is balanced in opinion when $b^I_{IJK} = \rho_{IJ}\rho_{JK}\rho_{KI} > 0$ and balanced in trust when $b^A_{IJK} = \frac{1}{2}(\eta_{IJ}\eta_{JK}\eta_{KI} + \eta_{JI}\eta_{IK}\eta_{KJ}) > 0$, and the aggregates

$$B_I = \left\langle b^I_{IJK} \right\rangle_{(IJK)}, \qquad B_A = \left\langle b^A_{IJK} \right\rangle_{(IJK)} \tag{7}$$

run over all $\binom{N}{3}$ triples. (The subscripts are historical, from the older names for the two sectors, and do not index an agent.) A random society has $B_I \approx B_A \approx 0$, half its triples frustrated; two internally coherent, mutually opposed blocs give $B_I = B_A = 1$ whatever the blocs are made of; and a society that cannot settle stays near zero or goes negative. Both aggregates follow in closed form from matrix products rather than by enumerating triples (Appendix J). Nothing in ρ, η or (7) refers to weights, gradients or K — only to a signed alignment and a signed trust per pair, which is what makes them measurable on any population of interacting agents.

The society splits in two. Figure 2 shows the two pairwise distributions before and after. Opinion overlap begins unimodal and centred on zero, of width 0.178 against the $1/\sqrt{K} = 0.183$ expected of random vectors in $K = 30$ dimensions, and ends bimodal with peaks at -0.59 and $+0.63$ and no pair left within 0.1 of zero. Trust begins spread across the interior and ends saturated, 35% of ordered pairs beyond $|\eta| > 0.99$ and a mean $|\eta|$ of 0.986: the split in trust is built by the dynamics, not supplied by the initial condition. Neither panel depends on how the agents are labelled, so neither can be an artefact of an ordering imposed on them.

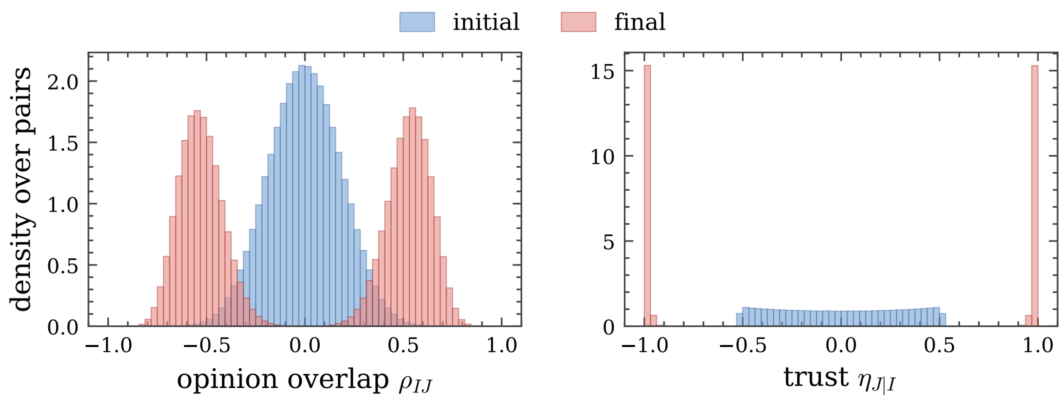

Figure 2: **An unbiased society splits in two.** Eight independent societies of $N = 400$ agents at $d = 0$, simple agenda, before and after the calibrated 500 interactions per ordered pair, with their pairs pooled: 638 400 unordered pairs for the opinion overlap and 1 276 800 ordered pairs for the trust. Blue is the initial state, red the final one. Neither panel depends on how the agents are labelled, so neither can be an artefact of sorting them. Opinion overlap starts unimodal at zero, of width $1/\sqrt{K}$ as random vectors in $K = 30$ dimensions require, and ends bimodal; trust starts spread across the interior and ends saturated at ± 1, so the split in trust is built by the dynamics and not supplied by the initial condition. Bimodality alone would show only that every pair has taken a side; that the sides are exactly *two* blocs is the sign-balance figure quoted in the text.

The two panels are also aligned with each other: agents end up trusting those who agree with them and distrusting those who do not, which is not imposed anywhere in §3 but follows from the signs of the modulation functions. Balance places the state. Trust balance reaches $B_A = 0.96$, so the trust network is almost perfectly organised, while opinion balance reaches only $B_I = 0.19$ — a gap that §4.2 explains. That the sides are exactly *two* rather than several is a separate question, since a bimodal distribution is equally consistent with three or four mutually hostile groups; counting triples by sign alone settles it, as a signed complete graph is balanced precisely when it splits into two mutually hostile cliques (Cartwright & Harary, 1956). Three equal groups would give $3/4$ and random signs $1/2$. Of the 84 694 400 triples in Figure 2, not one is unbalanced in trust, and in opinion the shortfall is 1 590 triples, two thousandths of one per cent, traceable to the few pairs whose overlap is so close to zero that its sign is arbitrary.

Two features of the split matter later. It is *spontaneous*: which agent lands in which faction is set by the initial condition, and the two are equal only up to fluctuation, splitting between $190/210$ and $212/188$ of 400 agents across the eight societies. And it is *class-blind*: the labels κ_I exist, but no part of the dynamics has consulted them, so the factions cut across them. What §5 adds is a mechanism for aligning this pre-existing split with the labels.

4.2 The agenda-size effect

Both balances rise from zero towards one, but not at the same rate, and which one leads is controlled by the complexity of the agenda,

$$\alpha = \frac{P}{K}, \tag{8}$$

the number of issues in play per dimension of the space they are represented in. Following a society through the (B_I, B_A) plane as it evolves, rather than measuring it once at the end, separates the two.

Figure 3 does this at $d = 0$ for agendas spanning four orders of magnitude in α. For a simple agenda the path runs up the B_A axis: trust structure forms almost completely before opinions begin to sort, ending at $(B_I, B_A) = (0.02, 0.97)$ for $\alpha = 0.03$. For a complex agenda the path stays below the diagonal, ending at $(0.97, 0.74)$ for $\alpha = 333$: opinions separate first and trust follows. The crossover is at $\alpha \approx 1.7$.

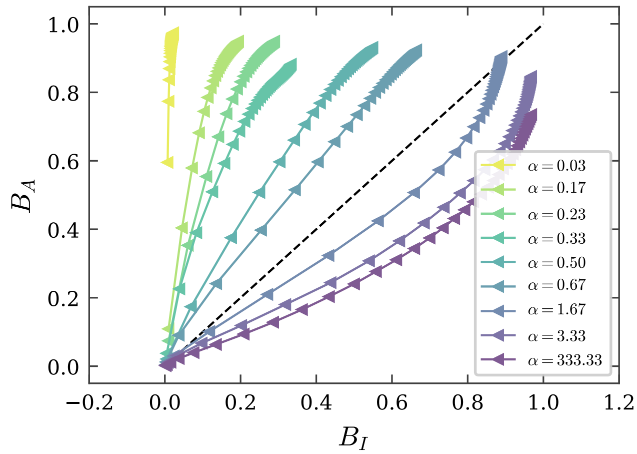

Figure 3: **Which polarisation comes first is set by the agenda.** Trajectories in the (B_I, B_A) plane with no discrimination ($d = 0$), one curve per agenda complexity $\alpha = P/K$, markers equally spaced in time — they crowd at the end because the annealed dynamics slows, as §3.1 requires. Simple agendas run above the diagonal (distrust forms first), complex ones below it (disagreement first), crossing over at $\alpha \approx 1.7$.

The asymmetry has an exact cause. Every weight update is along $C_r\hat{x}$, which stays in the span of the agenda for all time, so the component of each agent's weight vector orthogonal to that span is conserved by the dynamics (Appendix A). For $\alpha < 1$ each agent therefore keeps a frozen random component that no interaction can touch, which dilutes every overlap and so caps B_I — at 0.19 for $\alpha = 0.17$, against 0.95 for $\alpha = 3.3$. No amount of running closes that gap, because it is a property of the space and not of the time spent in it. Trust has no such ceiling: $\mu_{e|r}$ is one scalar per ordered pair rather than a projection of a high-dimensional vector, so η saturates at any α.

Read as a mechanism, this says that whether groups come to dislike each other because they disagree, or come to disagree because they dislike each other, is a question with a parameter attached rather than an answer: narrow agendas put affect first, broad ones put opinion first, and the two are regimes of one system rather than rival accounts. The same ratio will explain, in §6, why one of the two discriminatory phases exists only for narrow agendas.

5 Discrimination as an error of representation

We now let some agents make a specific, minimal mistake: when they receive a message, they extend the representation of the issue with a feature that carries no information about the issue at all, and depends only on *who sent it*. If that feature is correlated with the emitter's class label, the agent is a *discriminating* agent, and the resulting shift of its opinion field is the *discrimination field*:

$$h_w^D = h_w + D_{\kappa_r, \kappa_e}, \tag{9}$$

where D is a 2×2 matrix indexed by the classes of receiver and emitter. A fraction f_d of the society discriminates, drawn independently of class; the rest use (2) unmodified.

This is a familiar failure viewed from an unusual angle: irrelevant features degrade a classifier, and if one correlates with a protected attribute the errors become structured along it. Here the classifier is one agent among many, and the question is what those errors do to the *population*.

What the field does. By (6), the sign of perceived agreement decides whether trust grows or decays. A receiver that adds $+d$ to h_w reads its counterpart as more agreeable than it is and becomes more trusting: it is *tolerant*. A receiver that adds $-d$ manufactures disagreement that was not there and becomes more distrustful: it is *intolerant*. Throughout this paper $d > 0$ therefore denotes **tolerance towards in-group and intolerance towards out-group** emitters, and $d < 0$ the reverse — discrimination in favour of the other class.

Geometrically, the field displaces the separatrix of the learning flow to $h_\mu = h_w + D$ (Figure 8, Appendix D). The undisplaced flow has two symmetric consonant basins, “trust and agree” and “distrust and disagree”, and a society whose agents fall into them polarises into two camps whose membership has nothing to do with class. Displacing the separatrix breaks that symmetry: a receiver meeting an out-group emitter has an enlarged “distrust and disagree” basin, and one meeting an in-group emitter an enlarged “trust and agree” basin. Every discriminating agent is thus biased towards the same alignment of distrust with class, and once enough of them are, the whole society can lock into it.

Six ways to discriminate. The matrix D can be filled in six inequivalent ways, listed in Appendix G: one class may favour its own without hostility to the other, be hostile without favouritism, or both; and either one class or both may discriminate. All the phase diagrams below use the fully symmetric case

$$D \;=\; d\begin{pmatrix} +1 & -1 \\ -1 & +1 \end{pmatrix}, \tag{10}$$

in which both classes favour their own and are hostile to the other. It is the case in which the two classes play equivalent roles, so any asymmetry that appears in the results is spontaneous rather than imposed.

A note on the sign convention. An earlier version of this model (Anonymous, 2026) tabulates D with the opposite sign while placing the discriminatory phases at $d > 0$ in its text and figures; the two are inconsistent, and the inconsistency exchanges the halves of the phase diagram. Appendix H works this through and shows the maps under both readings. We use (9) with (10) as stated, which is the reading that agrees with the algorithm.

5.1 Three correlations

Balance says whether a society is organised; it does not say what it is organised around. Three pairwise correlations do, and they are what the phase diagram is built from. The first pairs the two sectors against each other,

$$R_{w,\mu} = \big\langle (\eta_{I|J} + \eta_{J|I})\, \rho_{IJ} \big\rangle, \tag{11}$$

asking whether agents trust those who agree with them; it is positive in the polarised state of §4, reaching $+0.56$ there from zero at initialisation. The other two refer to class. Writing $G_{IJ} = \kappa_I \kappa_J$ for the class indicator ($+1$ for same-class pairs, -1 otherwise),

$$R_{\mu,c} = \big\langle G_{IJ}(\eta_{I|J} + \eta_{J|I}) \big\rangle, \qquad R_{c,w} = \big\langle G_{IJ}\, \rho_{IJ} \big\rangle, \tag{12}$$

all three averaged over unordered pairs and normalised to $[-1, 1]$. $R_{\mu,c}$ is the order parameter of discrimination: it asks whether trust follows class, distinguishing a society that has merely polarised from one that has polarised *along the label*. $R_{c,w}$ asks whether opinion follows class, and separates the two discriminatory phases from each other — a society can have $R_{\mu,c}$ near one with $R_{c,w}$ near zero, and that combination is a distinct state rather than a degenerate case. Appendix I records why G is signed rather than $\{0, 1\}$ and how the three correlations are normalised.

6 The phase diagram

We simulate societies of $N = 40$ agents over a 200×200 grid in (d, f_d), one society per grid point, and measure the five order parameters of §4.1 and §5.1 after a fixed number of interactions, with both classes discriminating (case (10)). Parameters and their provenance are in Appendix K; each pixel is a single realisation, so the speckle is society-to-society fluctuation, not a rendering artefact.

Discrimination needs a quorum, and then arrives abruptly. At small f_d the society stays neutral no matter how strong the bias: the few discriminating agents are outvoted by the class-blind majority they learn from. The trust–class correlation at large d first exceeds 0.5 around $f_d \approx 0.4$ (0.43 for the simple agenda, 0.35 for the complex one), and past that the transition in d sharpens quickly: at $f_d = 1$ the band in which $|R_{\mu,c}| < 0.25$ is only 0.54 wide, and $R_{\mu,c}$ runs from -0.99 at $d = -1$ to $+0.99$ at $d = +1$. This is the first practical consequence of the model: the state of the population is not a smooth function of how biased its members are, and a minority of biased agents leaves no trace at all until it reaches a threshold fraction.

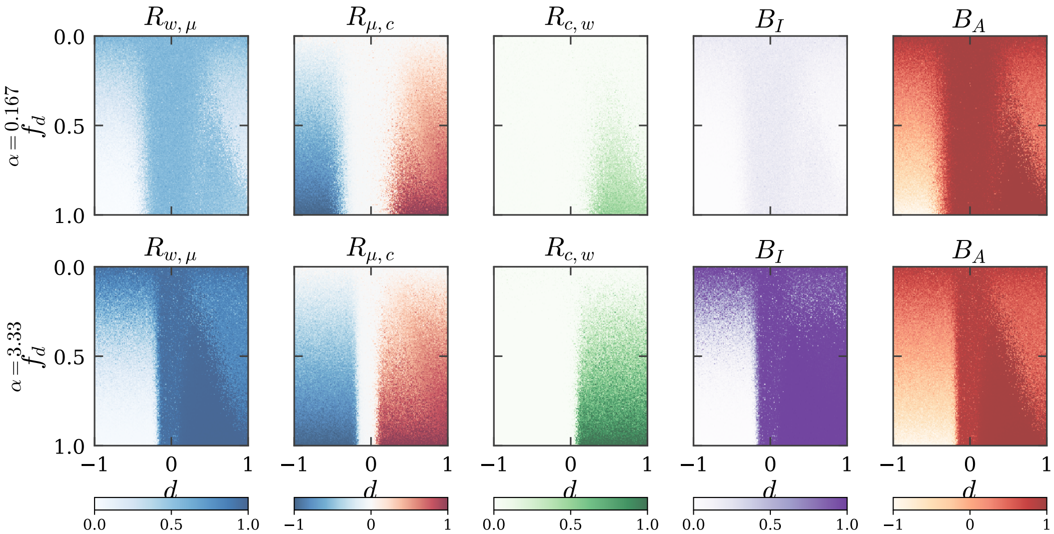

Figure 4: **All five order parameters over the (d, f_d) plane**, for a simple agenda ($\alpha = 0.17$, top) and a complex one ($\alpha = 3.3$, bottom). $R_{\mu,c}$ is the discrimination order parameter, flipping sign at $d \approx 0$ ever more sharply as f_d grows; $R_{c,w}$ is non-zero only for $d > 0$ and, for the simple agenda, *falls back* at the largest d, which is what separates the two discriminatory phases; B_I separates the agenda sizes, featureless above and saturated below; and B_A is the only panel in which the frustrated region at lower left announces itself, going strongly negative where everything else merely goes to zero.

Two discriminatory phases, and only the simple agenda has both. Both discriminatory regions have distrust aligned to class; they differ in whether opinion is aligned too. Along $f_d = 0.9$ in the top row of Figure 4, $R_{c,w}$ rises to 0.44 at $d \approx 0.6$ and then *falls* to 0.30 by $d = 1$, while $R_{\mu,c}$ keeps climbing, from 0.78 to 0.87 — a gap of 0.55 where the two started level. Opinion balance follows $R_{c,w}$ down, B_I dropping from 0.19 to 0.06 across the same row. At moderate bias the society splits into two camps that both distrust and disagree with each other (region III); at strong bias the disagreement becomes unnecessary and class membership alone carries the hostility (region IV), which is what the magenta corner of Figure 5 shows.

The complex agenda has no region IV. In the bottom row, $R_{\mu,c}$ and $R_{c,w}$ track each other across the whole $d > 0$ half — 0.84 against 0.85 at $d = 0.15$, 0.91 against 0.91 at $d = 0.75$, and still only 0.04 apart at $d = 1$ against the simple agenda's 0.55 — while B_I saturates at 0.98. When there are many issues to disagree about, the opinion split tracks the class split completely, so distrust and disagreement never come apart. Class-only hostility requires the opposite: an agenda narrow enough that most of every opinion lies in directions no issue probes, which by §4 caps $R_{c,w}$ however long the society runs. Region (IV) is therefore not a slow approach to region (III) but a distinct state, available only below $\alpha \approx 1$.

Favouring the out-group produces frustration, not inverted discrimination. For $d < 0$ the correlations are the sign-mirror of the $d > 0$ side, which invites reading the left half of the diagram as "reverse discrimination". The balance measures say otherwise (Figure 4). In the discriminatory region $B_A = +0.97$: nearly every triple is balanced in trust, as two coherent mutually hostile blocs require. At $d = -1$, $f_d = 0.9$, $B_A = -0.71$ for both agendas — strongly *anti*-balanced — with $B_I = 0.00$, and $R_{w,\mu}$ collapses to 0.03 — the only place it does. No consistent assignment of trust exists: each agent is pulled towards the other class while its neighbours are pulled towards theirs, so the society settles into a spin glass rather than a mirror image. Pair correlations alone would report the region as simply inverted; the triple-level measurement identifies it as disordered.

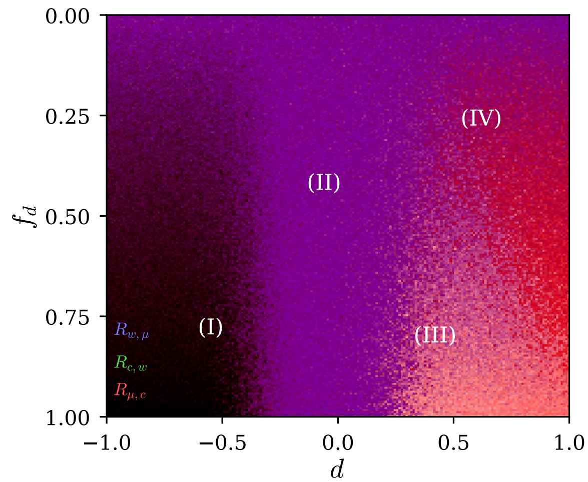

Figure 5: **Four collective states.** The three correlations of Figure 4 (simple agenda) composited as colour channels — red $R_{\mu,c}$, green $R_{c,w}$, blue $R_{w,\mu}$ — so that colour names the phase: (I) dark, frustrated reverse discrimination; (II) blue-violet, neutral polarisation unrelated to class; (III) pale, opinion tracking class; (IV) magenta, class alone.

7 Discussion

Optimality is context-dependent, and the context is group size. The rule these agents use is optimal for the problem it was derived for: one student, one teacher, a noisy channel whose reliability must be inferred online. Nothing about it is careless. Yet the same rule, run in a large population with one class-correlated perturbation, converts an informationally empty label into a stable architecture of group distrust. The mechanism is specific: it acts through F_μ, and a rule with no trust sector has none to bias — under a plain Hebbian update the discrimination field does nothing at all, since the update does not depend on h_w. We argue this rather than measure it; the ablation is cheap and worth running. What exposes the population is precisely the machinery that makes each agent good at inference, that it draws conclusions about *sources* as well as about the world. Giving interacting agents a well-founded model of each other's reliability is not a neutral improvement.

Individually small, collectively discontinuous. A single discriminating agent is biased by $O(d)$; the population's state is not, moving between phases over a narrow interval in d once f_d passes its threshold. A protocol that inspects agents one at a time — the standard protocol — can therefore certify every agent as approximately unbiased while the population sits in a discriminatory phase. Auditing a multi-agent system needs population-level order parameters, and the ones used here cost two elicited matrices.

Two results that reframe standard questions. The reversal at $\alpha \approx 1.7$ makes "do groups disagree because they dislike each other, or the reverse" a question about a parameter rather than a choice between theories. And making d negative, so agents favour the out-group, gives not a tidy inverted society but frustration: $B_A < 0$ with $B_I \approx 0$, order parameters near zero not because nothing is happening but because nothing can settle. Pair correlations alone would misreport that state as neutral; only the balance measures identify it as a spin glass.

Limitations. The agents are single-layer perceptrons over a fixed shared embedding, all pairs may interact, and the society is closed and of fixed size. The dynamics anneals rather than reaching a stationary state, so our phases are phases at a finite interaction count, and the class labels are exactly inert — the clean version of a case that is never clean. Most importantly the result rests on one substrate, agents whose learning rule we can write down; whether the same phases appear where the agents are not analysable is the obvious next question, which the order parameters of §4.1 were built to make answerable but which we do not answer. Human societies are a further step again.

Reproducibility statement

Every figure in this paper is produced by a script in the accompanying code, which contains the model, the order parameters, and the sweep driver, together with a test suite that checks the modulation functions against finite differences of $\log Z$, the invariants of the dynamics (the covariance stays symmetric positive-definite, the trust variance stays positive), and the order parameters against their known limits. One command regenerates all figures at a choice of three resolutions; sweeps are cached, so a figure can be restyled without re-simulating.

The model has few parameters and we state all of them, including the ones the formulation leaves open. §6 gives the values used; Appendix K gives the full table, separating quantities fixed by the model from those we chose, and records how each choice was made and what happens under alternatives. Two conventions where a literal reading of the model as previously written is ambiguous or inconsistent — the sign of the discrimination field, and the normalisation of the class indicator — are resolved explicitly in Appendices H and I, with both readings implemented behind a flag so either can be reproduced.

Ethics statement

This paper studies how discriminatory structure can arise among interacting learning agents. Two points bear stating.

The model is not a claim about people. Its agents are single-layer perceptrons, its groups are arbitrary inert labels, and its "tolerance" is a single scalar. We use the vocabulary of in-group and out-group because it names the structure compactly, not because the model licenses inferences about human prejudice, whose causes are historical, institutional and material in ways nothing here represents. Where we discuss the trust-versus-opinion polarisation literature, we do so to point out that the model identifies a parameter that selects between two regimes, not to adjudicate an empirical question about any actual polity.

The practical direction of the work is defensive. The result implies that a multi-agent system can occupy a discriminatory collective state while every agent in it passes an individual audit, and that the transition into that state is sharp in the strength of a per-agent bias. That argues for population-level measurement of deployed multi-agent systems, and the order parameters used here are inexpensive enough to serve as one.

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A The agenda's orthogonal complement is conserved

Let $V_P = \operatorname{span}\{\hat{x}_1, \dots, \hat{x}_P\}$ be the subspace spanned by the agenda. *Claim:* for every agent, the component of $\hat{w}$ orthogonal to V_P is unchanged by (4).

By (4) the weight update is a multiple of $C_r\hat{x}$ with $\hat{x} \in V_P$, so it is enough that $C\hat{x} \in V_P$ at all times. At initialisation $C = I$ and $C\hat{x} = \hat{x} \in V_P$. Suppose $C_t\hat{x}_i \in V_P$ for every issue. The covariance update is $C_{t+1} = C_t + a\,(C_t\hat{x})(C_t\hat{x})^\top$ for the issue $\hat{x}$ just seen, so for any issue $\hat{y}$,

$$C_{t+1}\hat{y} \;=\; C_t\hat{y} \;+\; a\,(C_t\hat{x})\,\big(C_t\hat{x}\cdot\hat{y}\big), \tag{13}$$

in which the first term lies in V_P by hypothesis and the second is a multiple of $C_t\hat{x} \in V_P$. Hence $C_t\hat{x}_i \in V_P$ for all t by induction, every increment to $\hat{w}$ lies in V_P, and the orthogonal component is conserved. □

The conservation is exact in the simulations: in the run of Figure 2, $\|w_\perp\| = 4.956$ at initialisation and 4.956 after $780\,000$ interactions, and the test suite asserts it numerically. It is a property of the *agenda*, not of the discrimination field: it holds at every d and f_d.

The ceiling it puts on opinion overlap. Split $w_I = u_I + n_I$ into the learned part inside V_P and the frozen part outside it. Learning aligns the u_I while the n_I stay random and mutually almost orthogonal, so

$$\rho_{IJ} \;\approx\; \pm f_I f_J, \qquad f_I = \frac{\|u_I\|}{\|w_I\|}, \tag{14}$$

and opinion overlap is diluted by whatever fraction of each weight vector learning cannot reach. Two facts fix that fraction. For $\alpha < 1$ a $(1-\alpha)$ fraction of directions is untouched, so $n_I \neq 0$; and because the dynamics anneals — $C \to 0$, so $\|u_I\|$ stops growing — f settles below one instead of creeping towards it. Measured across four orders of magnitude in α:

$\alpha = P/K$	0.03	0.17	0.50	1	3.3
f^2, predicted by (14)	0.29	0.62	0.84	1.00	1.00
$\langle\lvert\rho\rvert\rangle$, measured	0.30	0.62	0.82	0.93	0.99

For $\alpha < 1$ the prediction holds to 0.02. For $\alpha \geq 1$ the agenda spans $\mathbb{R}^K$, the complement is empty, and the ceiling disappears, the residual shortfall being the finite number of interactions rather than a conserved quantity.

B All four modulation functions

C Where blame for a surprise falls

D The learning flow under a discrimination field

E Derivation of the learning algorithm

The receiver's belief before an interaction is the product of Gaussians (1). It observes σ_e, the emitter's opinion on issue $\hat{x}$, and treats it as a report of an unknown true label σ_T transmitted through a channel that flips the label with probability $\varepsilon(z)$. Marginalising the joint distribution of labels and of the issue as the emitter may have parsed it, y_t,

$$L(\sigma_e \mid \hat{x}, u) \;=\; \sum_{\sigma_T} \int P(\sigma_e \mid \sigma_T, y_t, \hat{x}, u)\, P(\sigma_T \mid y_t, \hat{x}, u)\, P(y_t \mid \hat{x}, u)\, \mathrm{d}y_t, \tag{15}$$

with $P(\sigma_e \mid \sigma_T, z) = (1-\varepsilon(z))\delta_{\sigma_e,\sigma_T} + \varepsilon(z)\delta_{\sigma_e,-\sigma_T}$ and $P(\sigma_T \mid y_t, w) = \Theta(\sigma_T\, y_t \cdot w)$. Two modelling choices keep the algebra Gaussian. Taking $\varepsilon(z) = \Phi(z)$ with z Gaussian of mean $\mu_{e|r}$ and variance $V_{e|r}$ makes the flip probability a bounded function of a Gaussian variable while preserving conjugacy of the trust sector — a Beta distribution for ε, the tempting alternative, does not. Taking $P(y_t \mid \hat{x}, u)$ Gaussian with variance $v_p = \|w\|^{-2}$ lets an inexperienced agent, whose weights are small, be correspondingly unsure that it has parsed the issue as the emitter did. Together these give the likelihood

$$L(\sigma_e \mid z, \hat{x}, w) \;=\; \Phi(z)\,\Phi(-w \cdot \hat{x}\sigma_e) + \big(1 - \Phi(z)\big)\,\Phi(w \cdot \hat{x}\sigma_e), \tag{16}$$

and the covariance stays block diagonal in the two sectors if it starts so.

Averaging the likelihood over the current belief gives the evidence

$$Z(\sigma_e \mid \hat{x}, D_t) \;=\; \mathbb{E}_{Q_t}\big[L(\sigma_e \mid z, u, \hat{x})\big] \;=\; \sum_{\sigma_T} \mathbb{E}_z\big[P(\sigma_e \mid \sigma_T, \hat{x}, z)\big]\, \mathbb{E}_w\big[\Phi(\sigma_T\, w \cdot \hat{x})\big], \tag{17}$$

and the Gaussian integrals collapse onto the two scaled fields (2), giving (3): the evidence depends on the receiver's state only through (h_w, h_μ). This is the symmetry that makes the two sectors mirror images.

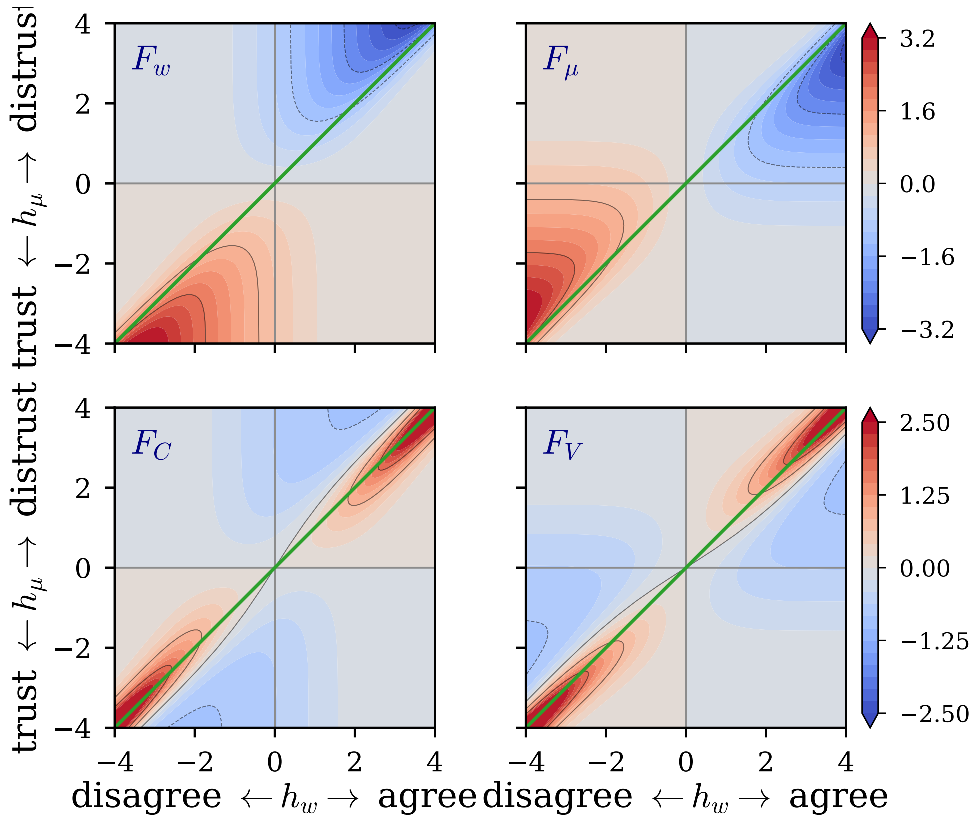

Figure 6: The two functions that move the means, F_w and F_μ, above — the pair shown in Figure 1 — and the two that anneal the uncertainties, F_C and F_V, below. Each row shares a range. The mirror symmetry across $h_\mu = h_w$ holds in both rows: $F_w(x, y) = F_\mu(y, x)$ and $F_C(x, y) = F_V(y, x)$. F_C and F_V are negative almost everywhere, which is why both uncertainties shrink monotonically and the dynamics anneals (§3.1).

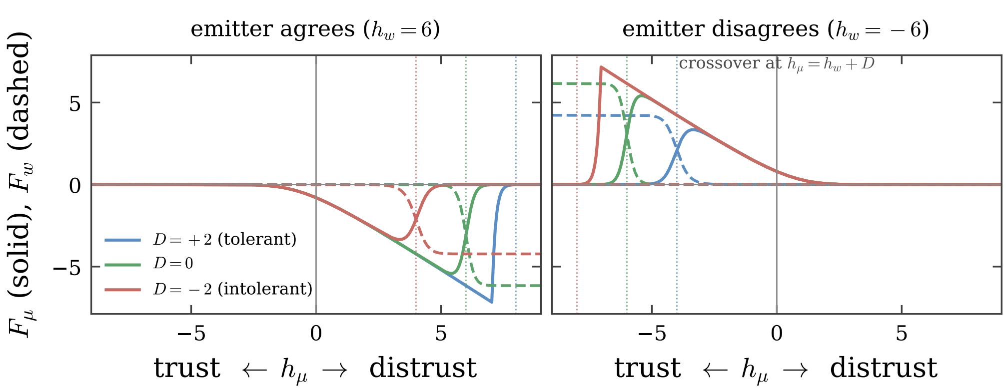

Figure 7: Cuts through the modulation functions at fixed opinion field, $h_w = \pm 6$. Below the crossover at $h_\mu = h_w + D$ the trust sector moves (solid, F_μ) and the receiver revises its trust; above it the opinion sector moves (dashed, F_w) and the receiver unlearns what the emitter said. The discrimination field D displaces the crossover — dotted verticals, one per value of D — which is the entire mechanism studied in this paper, visible in a single coordinate.

The exact Bayesian posterior is not Gaussian. Replacing it by the Gaussian of maximum entropy relative to the prior, subject to matching the first two moments of the posterior, gives updates in

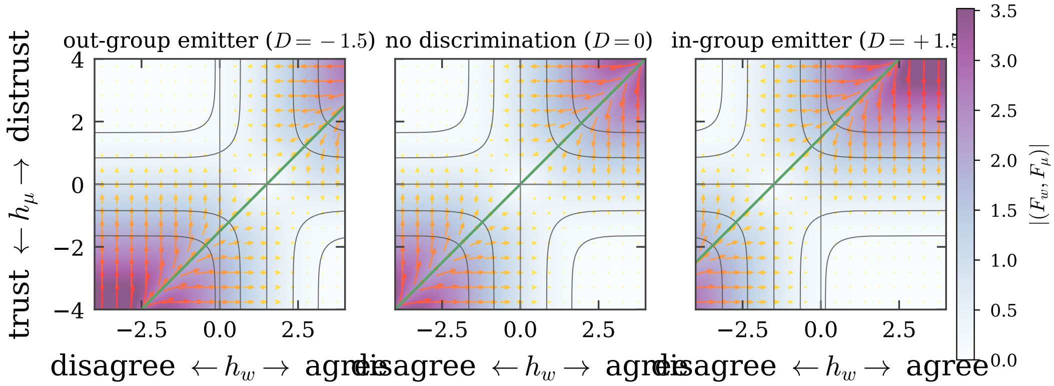

Figure 8: **The discrimination field bends the flow of learning.** Arrows are the change (F_w, F_μ) that one interaction induces in the receiver's fields; the background is $|(F_w, F_\mu)|$ and the thin contours are the evidence Z. Centre: without discrimination the flow escapes the two dissonant quadrants into two symmetric consonant basins, separated by $h_\mu = h_w$ (green). Left and right: a discriminating receiver evaluates the flow at $h_w + D$, which moves the separatrix and enlarges one basin at the other's expense — "distrust and disagree" for an out-group emitter ($D = -d$), "trust and agree" for an in-group one ($D = +d$).

terms of derivatives of the log evidence,

$$\hat{w}_{t+1} = \hat{w}_t + C_t \frac{\partial \ln Z}{\partial \hat{w}}, \qquad C_{t+1} = C_t + C_t \frac{\partial^2 \ln Z}{\partial \hat{w}\, \partial \hat{w}^\top} C_t, \tag{18}$$

and identically in the trust sector with $\mu_{e|r}, V_{e|r}$ in place of $\hat{w}, C$. Expressing the derivatives in the scaled fields yields (4)–(5) with the modulation functions (6). The first is a Hebbian term $\sigma_e \hat{x}$ with a tensorial, self-annealing rate; the rest is the online estimation of the channel.

F Genealogy of the learning rule

The update (4)–(5) descends from the variational optimisation of modulated Hebbian on-line learning (Kinouchi & Caticha, 1992; Kinzel & Ruján, 1990), given a Bayesian reading by Opper (1996; 1998) and applied across architectures by Solla & Winther (1999); the entropic-dynamics formulation we use is Caticha (2020). Learning through a noisy channel whose noise level must itself be inferred was analysed by Kinouchi & Caticha (1993) and Biehl et al. (1995); Alves & Caticha (2016) identified that inferred level with distrust and made it dynamical, with applications to voting (Caticha et al., 2015) and to panels of appellate judges (Caticha & Alves, 2019), and a mean-field treatment by Simões & Caticha (2018). The blame-attribution behaviour resembles the least-action rule of Mitchison & Durbin (1989), which suffers retarded learning (Kabashima, 1994), whereas the optimised algorithm attains generalisation error decaying as α^{-1} even under output noise (Simonetti & Caticha, 1996).

G The six discrimination cases

The matrix D in (9) admits six inequivalent fillings, where $+d$ is the tolerant entry and $-d$ the intolerant one (§5 and Appendix H). Rows index the receiver's class, columns the emitter's.

All phase diagrams in the main text use case 6, in which the two classes play symmetric roles.

H The sign of the discrimination field

The sign of D is fixed by the algorithm and not free. By (6), $F_\mu = (1 - 2\Phi(h_w))g(h_\mu)/Z$ is negative when $h_w > 0$, so perceived agreement drives distrust μ *down*. A receiver that adds $+d$ to h_w therefore becomes more trusting (tolerant) and one that adds $-d$ becomes more distrustful

case	D/d	who discriminates	how
1	$\begin{pmatrix} +1 & 0 \\ 0 & 0 \end{pmatrix}$	A only	in-group favouritism
2	$\begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix}$	A only	out-group hostility
3	$\begin{pmatrix} +1 & -1 \\ 0 & 0 \end{pmatrix}$	A only	both
4	$\begin{pmatrix} +1 & 0 \\ 0 & +1 \end{pmatrix}$	both	in-group favouritism
5	$\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$	both	out-group hostility
6	$\begin{pmatrix} +1 & -1 \\ -1 & +1 \end{pmatrix}$	both	both

(intolerant). Case 6 with $d > 0$ consequently gives in-group trust and out-group distrust, i.e. the discriminatory phase, which is the convention used throughout.

The companion manuscript (Anonymous, 2026) tabulates the in-group entry as $-d$ while its text and figures place the discriminatory phases at $d > 0$; those two statements are inconsistent, and taken literally the tabulated signs put the discriminatory phases at $d < 0$, mirroring every map in d. The same inconsistency appears in its description of the learning flow, where the panel labelled $D_{e|r} = -d$ is called the more tolerant one although $-d$ is the intolerant entry, and where the "distrust and disagree" basin is said to grow for $d > 0$ although adding $+d$ shrinks it. A single global sign reconciles everything; we adopt it in the form that leaves (9) untouched. The accompanying code implements both readings behind a flag, and Figure 9 runs the same sweep under each: the two rows are mirror images in d, and only the first matches the maps as published.

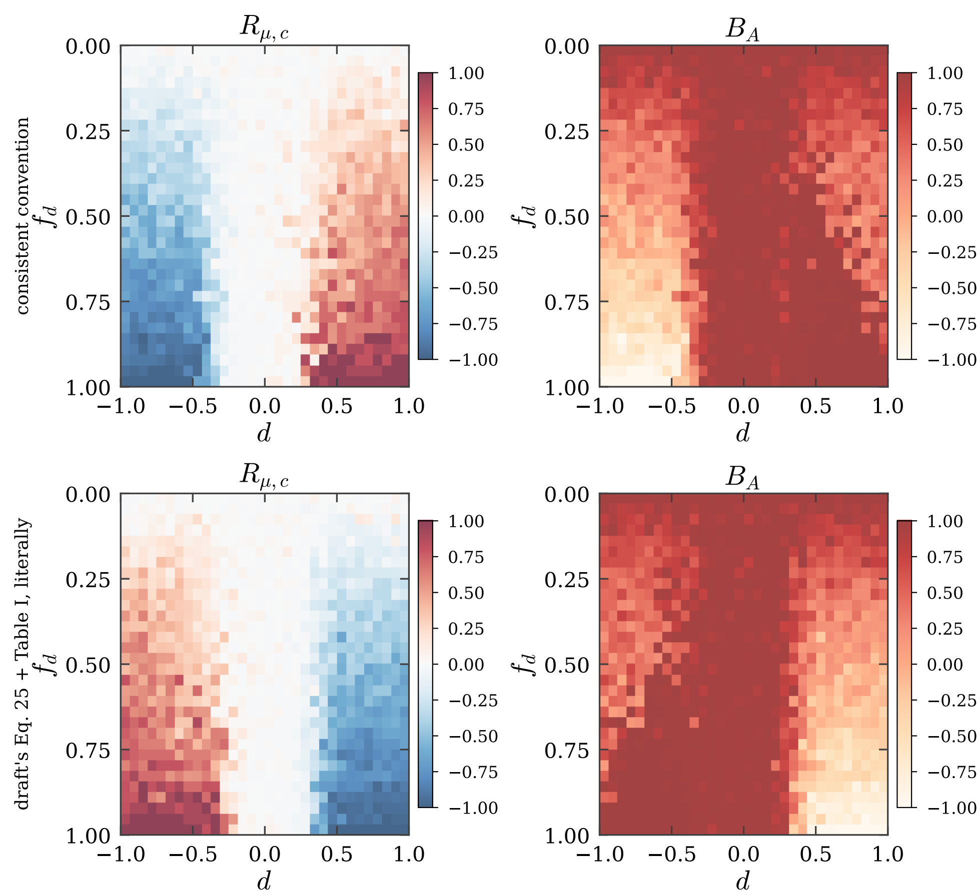

Figure 9: **The two readings of the discrimination field.** Trust–class correlation and trust balance over the (d, f_d) plane under the convention used in this paper (top) and under the companion manuscript's table read literally with its Eq. 25 (bottom). The physics is identical; which half of the d axis is discriminatory is not.

I CONVENTIONS IN THE ORDER PARAMETERS

Two choices in (12) deserve statement.

The class indicator. We use $G_{IJ} = \kappa_I \kappa_J \in \{-1, +1\}$. A $\{0, 1\}$ indicator, which counts only same-class pairs, cannot represent anti-correlation: under perfect reverse discrimination it returns $-(N/2-1)/(N-1)$ rather than -1, and its range depends on N. Since the reverse-discrimination half of the phase diagram is a substantive part of the result, the signed indicator is the appropriate one. Both are implemented.

Normalisation. $R_{w,\mu}$ and $R_{\mu,c}$ sum $\eta_{I|J} + \eta_{J|I}$ over unordered pairs, so dividing by $N(N-1)$ places them in $[-1, 1]$. $R_{c,w}$ has a single term per pair and the same divisor would cap it at $1/2$; we use $2/(N(N-1))$ so that all three share the range $[-1, 1]$ and the three colour channels of Figure 5 are commensurate.

J CLOSED FORM FOR THE BALANCE AGGREGATES

Enumerating triples costs $O(N^3)$ per society, which is prohibitive across a phase diagram. For any matrix M with unit diagonal — either ρ or η —

$$\sum_{\substack{(I,J,K)\\ \text{distinct, ordered}}} M_{IJ} M_{JK} M_{KI} \;=\; \mathrm{tr}(M^3) - 3\Big(\textstyle\sum_{I,J} M_{IJ} M_{JI} - N\Big) - N, \tag{19}$$

and each unordered triple appears six times, three as the forward cycle and three as the reverse, so $\langle b \rangle = \big[\text{right-hand side}\big] / \big(6\binom{N}{3}\big)$. For symmetric M this is B_I; for the asymmetric trust matrix the two cycle orientations are exactly the two terms of b^A, so the same expression gives B_A. Both are batched matrix products. The identity is verified against enumeration in the test suite.

K PARAMETERS

The model needs six numbers. One is recovered from the companion manuscript, one is calibrated against its published figures, two are prior conventions, and two are free choices; all six are listed with their provenance rather than left implicit, since the manuscript states none of them. The accompanying code exposes each of them and its calibration script reproduces the comparisons below.

symbol	meaning	value	provenance
K	dimension of the issue embedding	30	recovered (see below)
N	agents in the society	40	chosen; converged (see below)
P	issues on the agenda	5 / 100	chosen: $\alpha = P/K$ either side of 1
Δt	interactions per ordered pair	500	calibrated (see below)
C_0, V_0	initial uncertainties	$I, 1$	uninformative prior
μ_0	initial distrust	$\mathcal{U}(-1, 1)$	gives $B_I \approx B_A \approx 0$ at $t = 0$

K is recovered, not chosen. The companion manuscript reports balance trajectories at $\alpha \in \{0.03, 0.17, 0.23, 0.33, 0.50, 0.67, 1.67, 3.33, 333.33\}$. These are exactly $P/30$ for $P \in \{1, 5, 7, 10, 15, 20, 50, 100, 10^4\}$, which fixes $K = 30$; the language-agent protocol's thirty issues agrees.

Δt is calibrated against the published trajectories. The dynamics anneals rather than reaching a stationary state, so the measurement time matters and the companion manuscript does not state it (its text carries a literal placeholder in place of the value). We scan Δt and compare the endpoints of all nine balance trajectories against the published ones:

Δt (interactions per ordered pair)	60	125	250	**500**	1000
rms distance to published endpoints	0.63	0.31	0.15	**0.13**	0.15

The minimum is flat-bottomed and sits at $\Delta t = 500$, which we use throughout. Seven of the nine endpoints then agree to within about 0.1, and B_I matches the three largest agendas almost exactly, but no single Δt reproduces all nine: two middle curves ($\alpha = 0.23$, 0.33) sit above ours, and the two largest agendas have a lower published B_A than we obtain at any Δt that fits the rest. Either the society size differs from ours or those middle curves, which overlap within a few pixels in the printed figure, cannot be digitised reliably; we cannot distinguish the two. Dropping the ambiguous readings does not move the selected value. All the Δt-independent checks pass: the above/below-diagonal ordering with its crossover at $\alpha \approx 1.7$, the sharp sign flip of $R_{\mu,c}$ at $d \approx 0$, and the $R_{c,w}$ wedge confined to large d and large f_d.

$N = 40$ is large enough. Repeating five representative points of the plane at $N \in \{20, 30, 40, 60\}$ with twelve realisations each, the order parameters at $N = 60$ differ from those at $N = 40$ by at most 0.02 ($R_{\mu,c}$, $R_{c,w}$), 0.03 (B_I) and 0.05 (B_A), while $N = 20$ differs by up to 0.24. The maps are therefore converged in N at $N = 40$, and the cost of going further is steep: total work grows as N^3.

Numerical safeguards. Three floors are applied, none of which the model prescribes: on the evidence Z, whose exact modulation functions diverge in the deep dissonance corners; on the trust variance V, which $F_V < 0$ drives monotonically towards zero; and on the covariance update, clipped to the boundary at which $C + a(C\hat{x})(C\hat{x})^\top$ ceases to be positive semi-definite. The last is the only one that can alter a trajectory, and it never triggered in the runs reported here.